In [27]:
```
import numpy as np
import pandas as pd
from sklearn.datasets import fetch_20newsgroups
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.metrics import accuracy_score,f1_score,classification_report
from sklearn.naive_bayes import MultinomialNB,BernoulliNB
```

In [28]:
```
df = pd.read_csv("/home/backup/ml_datasets/20_newsgroups.csv")
print(df.head())
print(df.isnull().sum())
```

```
                                                text  target  \
0  \n\nI am sure some bashers of Pens fans are pr...      10
1  My brother is in the market for a high-perform...       3
2  \n\n\n\n\tFinally you said what you dream abou...      17
3  \nThink!\n\nIt's the SCSI card doing the DMA t...       3
4  1)    I have an old Jasmine drive which I cann...       4

               target_name
0          rec.sport.hockey
1  comp.sys.ibm.pc.hardware
2     talk.politics.mideast
3  comp.sys.ibm.pc.hardware
4     comp.sys.mac.hardware
text           380
target           0
target_name      0
dtype: int64
```

In [29]:
```
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2,random_state=42)
```

In [30]:
```
categories=['rec.sport.hockey','comp.sys.ibm.pc.hardware','talk.politics.mideast']
data=fetch_20newsgroups(subset='all',categories=categories,remove=('headers','footer
X=data.data
y=data.target
```

In [31]:
```
vectorizer_multi=CountVectorizer(stop_words='english',max_features=5000)
X_train_multi=vectorizer_multi.fit_transform(X_train)
X_test_multi=vectorizer_multi.transform(X_test)

#Bernoulli
vectorizer_bern=CountVectorizer(stop_words='english',binary=True,max_features=5000)
X_train_bern=vectorizer_bern.fit_transform(X_train)
X_test_bern=vectorizer_bern.transform(X_test)
```

In [32]:
```
mnb=MultinomialNB()
mnb.fit(X_train_multi,y_train)
y_pred=mnb.predict(X_test_multi)
predictions=pd.DataFrame({'Actual':y_test,'Predicted':y_pred})
acc=accuracy_score(y_test,y_pred)
print("Accuracy:",acc)
print(classification_report(y_test,y_pred))
```

```
Accuracy: 0.9504273504273504
              precision    recall  f1-score   support

           0       0.95      0.98      0.97       195
```

```
           1       0.96      0.93      0.94       201
           2       0.94      0.94      0.94       189

    accuracy                           0.95       585
   macro avg       0.95      0.95      0.95       585
weighted avg       0.95      0.95      0.95       585
```

In [33]:

```
bnb=BernoulliNB()
bnb.fit(X_train_bern,y_train)
y_pred=bnb.predict(X_test_bern)
predictions=pd.DataFrame({'Actual':y_test,'Predicted':y_pred})
acc2=accuracy_score(y_test,y_pred)
print("Accuracy:",acc)
print(classification_report(y_test,y_pred))
```

```
Accuracy: 0.9504273504273504
              precision    recall  f1-score   support

           0       0.64      0.99      0.78       195
           1       0.99      0.80      0.89       201
           2       0.98      0.63      0.77       189

    accuracy                           0.81       585
   macro avg       0.87      0.81      0.81       585
weighted avg       0.87      0.81      0.81       585
```

In [ ]:

In [ ]: